ANALOG AND DIGITAL COMMUNICATION
EXPERIMENT NO:2
Characteristics Study of Side Band Power and Frequencies in AM Using MATLAB

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```
PROGRAM:clc;
clear;
close all;

%% ============================================================
% CHARACTERISTICS STUDY OF SIDEBAND POWER AND FREQUENCIES IN
AM
% USING MATLAB
% ============================================================

%% 1. INITIAL PARAMETERS

Fs = 1e6;            % Sampling frequency = 1 MHz
T  = 0.01;           % Simulation time = 10 ms

t = 0:1/Fs:T-1/Fs;

% Carrier and message parameters
fc = 100e3;          % Carrier frequency = 100 kHz
fm = 10e3;           % Message frequency = 10 kHz

Ac = 1;              % Carrier amplitude
Am = 1;              % Message amplitude

R = 50;              % Load resistance = 50 ohms

%% 2. MODULATION INDEX

mu = Am/Ac;

fprintf('\n');
fprintf('============================================================\n');
fprintf('      AM SIDEBAND POWER AND FREQUENCY STUDY\n');
fprintf('============================================================\n');

fprintf('\nBasic Parameters:\n');
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```
fprintf('Carrier Frequency      = %.2f kHz\n',fc/1000);
fprintf('Message Frequency      = %.2f kHz\n',fm/1000);
fprintf('Carrier Amplitude      = %.2f V\n',Ac);
fprintf('Message Amplitude      = %.2f V\n',Am);
fprintf('Load Resistance        = %.2f Ohm\n',R);
fprintf('Modulation Index       = %.2f\n',mu);

%% 3. MESSAGE SIGNAL

message = Am*cos(2*pi*fm*t);

%% 4. CARRIER SIGNAL

carrier = Ac*cos(2*pi*fc*t);

%% 5. AM SIGNAL

AM_signal = Ac*(1 + mu*cos(2*pi*fm*t)) ...
        .* cos(2*pi*fc*t);

%% ============================================================
% 6. TIME DOMAIN PLOTS
% ============================================================

% Message signal

figure;

plot(t*1000,message,'LineWidth',1.2);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Message Signal');

grid on;

xlim([0 1]);

% Carrier signal

figure;

plot(t*1000,carrier,'LineWidth',1);
```

```
xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Carrier Signal');

grid on;

xlim([0 0.2]);

% AM signal

figure;

plot(t*1000,AM_signal,'LineWidth',1);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Amplitude Modulated (AM) Signal');

grid on;

xlim([0 1]);

%% ================================================================
% 7. AM SPECTRUM
% ================================================================

N = length(AM_signal);

AM_fft = fft(AM_signal);

AM_fft_shifted = fftshift(AM_fft);

f = (-N/2:N/2-1)*(Fs/N);

AM_magnitude = abs(AM_fft_shifted)/N;

figure;

plot(f/1000,AM_magnitude,'LineWidth',1.2);

xlabel('Frequency (kHz)');
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```
ylabel('Magnitude');

title('AM Frequency Spectrum');

grid on;

xlim([70 130]);

%% ============================================================
% 8. FIND USB AND LSB FREQUENCIES
% ============================================================

f_USB = fc + fm;

f_LSB = fc - fm;

fprintf('\n');
fprintf('Sideband Frequencies:\n');

fprintf('LSB Frequency          = %.2f kHz\n',f_LSB/1000);

fprintf('Carrier Frequency       = %.2f kHz\n',fc/1000);

fprintf('USB Frequency          = %.2f kHz\n',f_USB/1000);

%% ============================================================
% 9. BANDWIDTH
% ============================================================

BW_AM = 2*fm;

fprintf('\n');
fprintf('AM Bandwidth:\n');

fprintf('BW = 2fm = %.2f kHz\n',BW_AM/1000);

%% ============================================================
% 10. CARRIER POWER
% ============================================================

Pc = Ac^2/(2*R);

fprintf('\n');
fprintf('Carrier Power:\n');
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fprintf('Pc = %.6f W\n',Pc);

%% ========================================================================
% 11. SIDEband POWER
% ========================================================================

P_USB = (mu^2 * Pc)/4;

P_LSB = (mu^2 * Pc)/4;

P_SB_total = P_USB + P_LSB;

%% ========================================================================
% 12. TOTAL AM POWER
% ========================================================================

P_total = Pc + P_SB_total;

%% ========================================================================
% 13. POWER EFFICIENCY
% ========================================================================

efficiency = (P_SB_total/P_total)*100;

%% ========================================================================
% 14. DISPLAY POWER RESULTS
% ========================================================================

fprintf('\n');
fprintf('Power Calculations:\n');

fprintf('Carrier Power           = %.6f W\n',Pc);

fprintf('USB Power               = %.6f W\n',P_USB);

fprintf('LSB Power               = %.6f W\n',P_LSB);

fprintf('Total Sideband Power    = %.6f W\n',P_SB_total);

fprintf('Total AM Power          = %.6f W\n',P_total);

fprintf('Transmission Efficiency  = %.2f %%\n',efficiency);
```

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%% ========================================================
% 15. MODULATION INDEX STUDY
% ========================================================

mu_values = 0.1:0.1:1.0;

P_USB_values = zeros(size(mu_values));

P_LSB_values = zeros(size(mu_values));

P_SB_values = zeros(size(mu_values));

P_total_values = zeros(size(mu_values));

Efficiency_values = zeros(size(mu_values));

for k = 1:length(mu_values)

   m_index = mu_values(k);

   P_USB_values(k) = ...
      (m_index^2 * Pc)/4;

   P_LSB_values(k) = ...
      (m_index^2 * Pc)/4;

   P_SB_values(k) = ...
      P_USB_values(k) + P_LSB_values(k);

   P_total_values(k) = ...
      Pc + P_SB_values(k);

   Efficiency_values(k) = ...
      (P_SB_values(k)/P_total_values(k))*100;

end

%% ========================================================
% 16. DISPLAY MODULATION INDEX TABLE
% ========================================================

fprintf('\n');
fprintf('=================================================\n');
fprintf('      POWER VS MODULATION INDEX\n');
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fprintf('=================================================\n');

fprintf('\n');

fprintf(' Mu     USB Power    LSB Power    Total SB     Total AM\n');

fprintf('-------------------------------------------------\n');

for k = 1:length(mu_values)

   fprintf(' %.1f     %.6f     %.6f     %.6f     %.6f\n', ...
      mu_values(k), ...
      P_USB_values(k), ...
      P_LSB_values(k), ...
      P_SB_values(k), ...
      P_total_values(k));

end

fprintf('-------------------------------------------------\n');

%% ================================================================
% 17. USB POWER VS MODULATION INDEX
% ================================================================

figure;

plot(mu_values,P_USB_values,'o-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('USB Power (W)');

title('USB Power vs Modulation Index');

grid on;

%% ================================================================
% 18. LSB POWER VS MODULATION INDEX
% ================================================================

figure;

plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);
```

```
xlabel('Modulation Index (\mu)');
ylabel('LSB Power (W)');

title('LSB Power vs Modulation Index');

grid on;

%% ============================================================
% 19. TOTAL SIDEBAND POWER VS MODULATION INDEX
% ============================================================

figure;

plot(mu_values,P_SB_values,'^-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Total Sideband Power (W)');

title('Total Sideband Power vs Modulation Index');

grid on;

%% ============================================================
% 20. TOTAL AM POWER VS MODULATION INDEX
% ============================================================

figure;

plot(mu_values,P_total_values,'o-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Total AM Power (W)');

title('Total AM Power vs Modulation Index');

grid on;

%% ============================================================
% 21. TRANSMISSION EFFICIENCY
% ============================================================

figure;

plot(mu_values,Efficiency_values,'d-','LineWidth',1.5);
```

```
xlabel('Modulation Index (\mu)');
ylabel('Efficiency (%)');

title('AM Transmission Efficiency vs Modulation Index');

grid on;

%% ============================================================
% 22. COMPARE USB AND LSB POWER
% ============================================================

figure;

plot(mu_values,P_USB_values,'o-','LineWidth',1.5);

hold on;

plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Power (W)');

title('USB and LSB Power Comparison');

legend('USB Power','LSB Power');

grid on;

%% ============================================================
% 23. AM SPECTRUM FOR DIFFERENT MODULATION INDICES
% ============================================================

mu_plot = [0.25 0.5 0.75 1.0];

figure;

for k = 1:length(mu_plot)

   mu_temp = mu_plot(k);

   AM_temp = Ac*(1 + ...
      mu_temp*cos(2*pi*fm*t)) ...
      .* cos(2*pi*fc*t);
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```
    spectrum_temp = fftshift(fft(AM_temp));

    magnitude_temp = ...
        abs(spectrum_temp)/N;

    subplot(4,1,k);

    plot(f/1000,magnitude_temp);

    xlabel('Frequency (kHz)');
    ylabel('Magnitude');

    title(['AM Spectrum, \mu = ', ...
        num2str(mu_temp)]);

    grid on;

    xlim([70 130]);

end

%% ================================================================
% 24. ENVELOPE OF AM SIGNAL
% ================================================================

envelope_upper = Ac*(1 + mu*cos(2*pi*fm*t));

envelope_lower = -Ac*(1 + mu*cos(2*pi*fm*t));

figure;

plot(t*1000,AM_signal);

hold on;

plot(t*1000,envelope_upper,'--','LineWidth',1.2);

plot(t*1000,envelope_lower,'--','LineWidth',1.2);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('AM Signal and Its Envelope');
```

```
legend('AM Signal','Upper Envelope','Lower Envelope');

grid on;

xlim([0 1]);

%% ============================================================
% 25. IMPORTANT RESULTS AT 100% MODULATION
% ============================================================

mu_100 = 1;

P_USB_100 = (mu_100^2 * Pc)/4;

P_LSB_100 = (mu_100^2 * Pc)/4;

P_SB_100 = P_USB_100 + P_LSB_100;

P_total_100 = Pc + P_SB_100;

efficiency_100 = ...
   (P_SB_100/P_total_100)*100;

fprintf('\n');
fprintf('============================================================\n');
fprintf('         RESULTS FOR 100%% MODULATION\n');
fprintf('============================================================\n');

fprintf('Carrier Power           = %.6f W\n',Pc);

fprintf('USB Power               = %.6f W\n',P_USB_100);

fprintf('LSB Power               = %.6f W\n',P_LSB_100);

fprintf('Total Sideband Power    = %.6f W\n',P_SB_100);

fprintf('Total AM Power          = %.6f W\n',P_total_100);

fprintf('Transmission Efficiency  = %.2f %%\n',efficiency_100);

fprintf('============================================================\n');

%% ============================================================
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% 26. FINAL CONCLUSION
% ========================================================================

fprintf('\n');
fprintf('CONCLUSION:\n');

fprintf(['1. USB frequency = fc + fm.\n']);

fprintf(['2. LSB frequency = fc - fm.\n']);

fprintf(['3. AM bandwidth = 2fm.\n']);

fprintf(['4. Sideband power increases with mu^2.\n']);

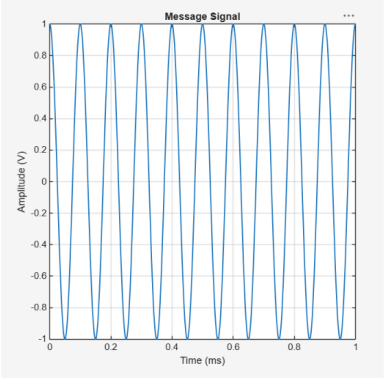
fprintf(['5. At 100%% modulation, total sideband power = Pc/2.\n']);

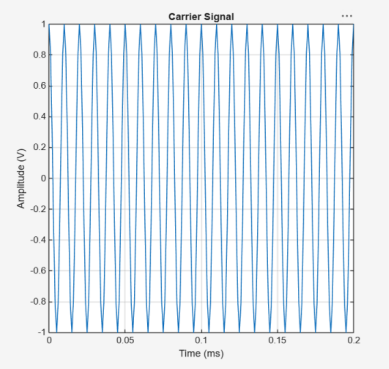
fprintf(['6. At 100%% modulation, total AM power = 1.5Pc.\n']);

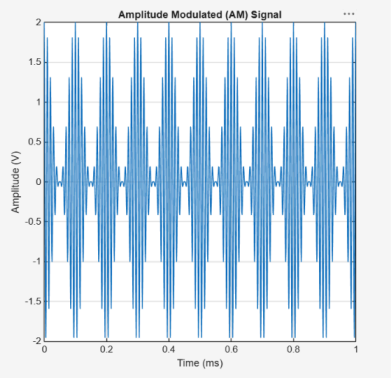
fprintf(['7. Maximum theoretical AM efficiency = 33.33%%.\n']);

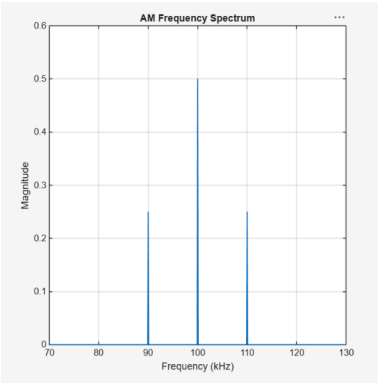
fprintf('\n========================================================\n');
fprintf('            SIMULATION COMPLETED\n');
fprintf('========================================================\n');
```

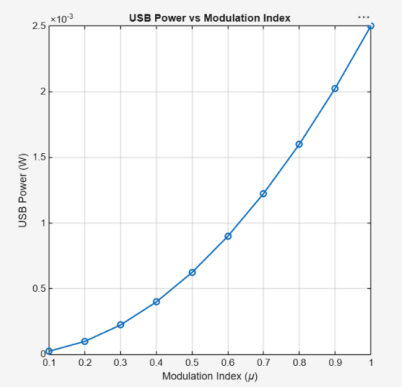
OUTPUTS:

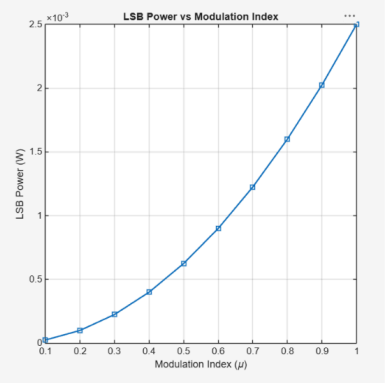


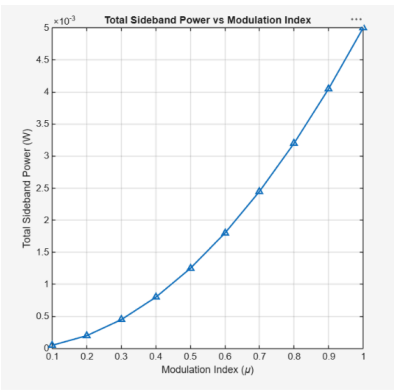


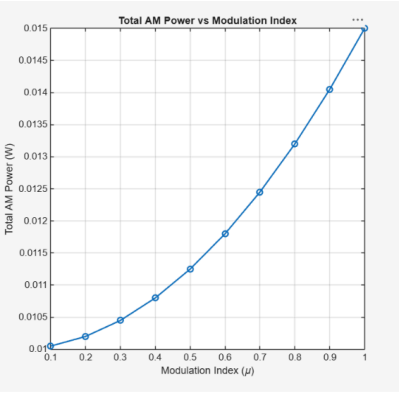


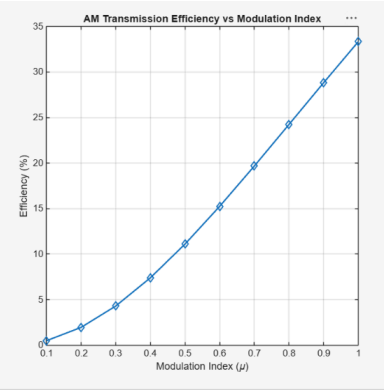


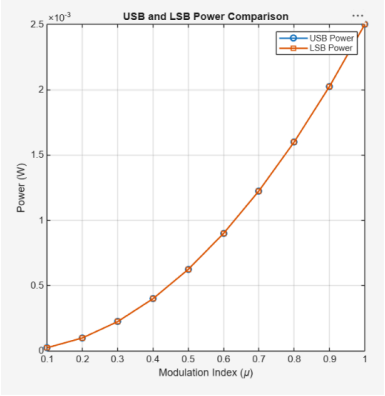


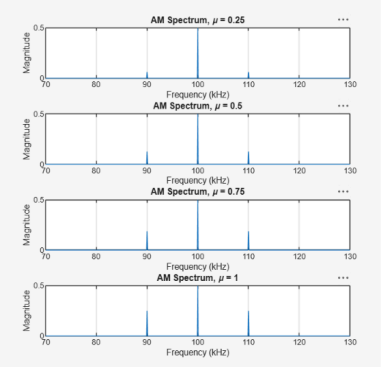


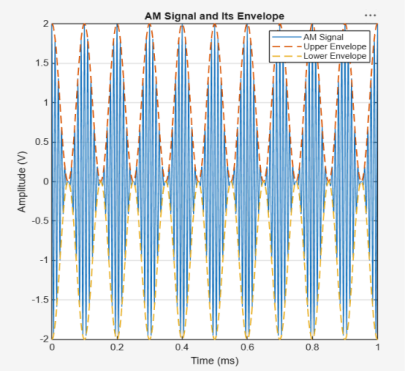

Total Sideband Power vs Modulation Index
×10^-3
Total Sideband Power (W)
Modulation Index (μ)


Total AM Power vs Modulation Index
Total AM Power (W)
Modulation Index (μ)


AM Transmission Efficiency vs Modulation Index
Efficiency (%)
Modulation Index (μ)


USB and LSB Power Comparison
×10^-3
USB Power
LSB Power
Power (W)
Modulation Index (μ)


AM Spectrum, μ = 0.25
AM Spectrum, μ = 0.5
AM Spectrum, μ = 0.75
AM Spectrum, μ = 1
Magnitude
Frequency (kHz)


AM Signal and Its Envelope
AM Signal
Upper Envelope
Lower Envelope
Amplitude (V)
Time (ms)